

CLASH OF THE **CLAWS**

RIT

**DebtMap: Migration Advisor
From Outdated to Up-To-
Date**

**Productivity platforms with
GenAI Integration**

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Problem Statement

We selected Productivity with GenAI because software maintenance and API migrations slow down development workflows. DebtMap uses lightweight static analysis with AI-assisted migration guidance to reduce developer effort and improve productivity.

Developers often spend significant effort manually reading migration guides, understanding updated documentation, and rewriting code. Existing tools can identify outdated APIs, but they do not provide actionable migration support.

Who Benefits ?

Individual Developers – Faster upgrades and reduced debugging effort
Engineering Teams – Better sprint planning and modernization visibility
Maintenance Engineers – Easier technical debt reduction at scale

Why It Matters

- Improves migration safety and reduces regressions
- Saves engineering time and maintenance cost
- Encourages secure, up-to-date dependencies
- Increases developer productivity and delivery speed

Current Solution & Gaps

	Tool	What it does	Gap
1.	Linters / Static Analyzers	Detect deprecated or bad code	Do not provide fixes or migration steps
2.	Security Tools (Snyk / CodeQL)	Identify outdated or vulnerable libraries	Do not show how to update or rewrite the code
3.	ChatGPT / AI Assistants	Explain APIs and concepts	No awareness of your specific codebase → gives generic answers
4.	Copilot / Code Generators	Help write new code	Cannot upgrade or refactor existing code

Your Solutions

What We Built

- Built a working AI-powered migration advisor
- Scans GitHub repositories for deprecated APIs
- Structured AI agent architecture inspired by OpenClaw concepts
- Generates migration explanations, upgrade steps, and rewritten code
- Connected frontend, backend, scanner, and AI pipeline

Core Idea

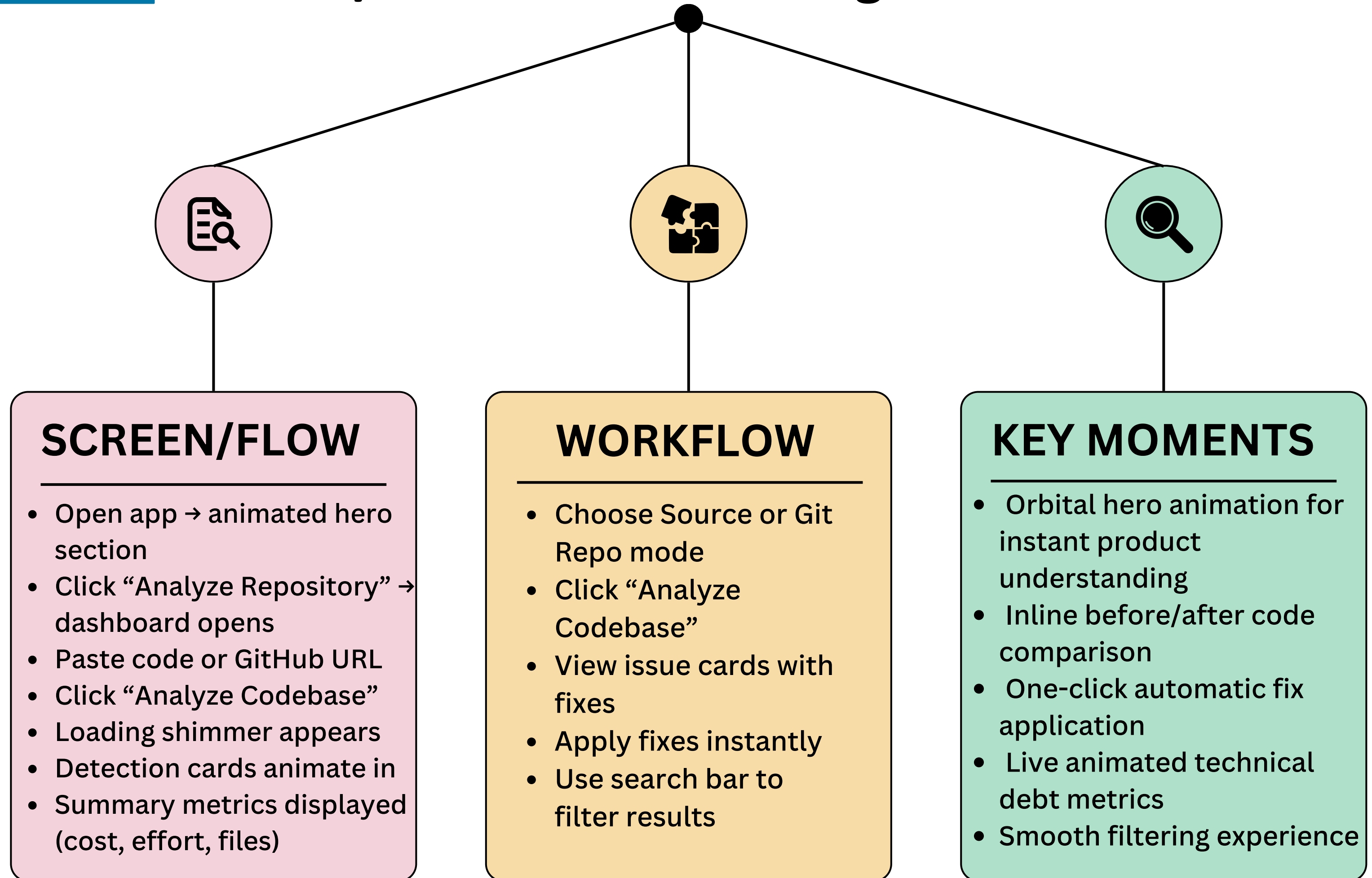
Current developer tools can detect deprecated APIs but usually stop at showing warnings. Developers need to manually understand the issue, search documentation, and rewrite code .

DebtMap reduces this migration effort by combining repository scanning with AI-generated migration guidance. The system detects deprecated code, explains why it is outdated, and suggests updated implementations directly in the workflow.

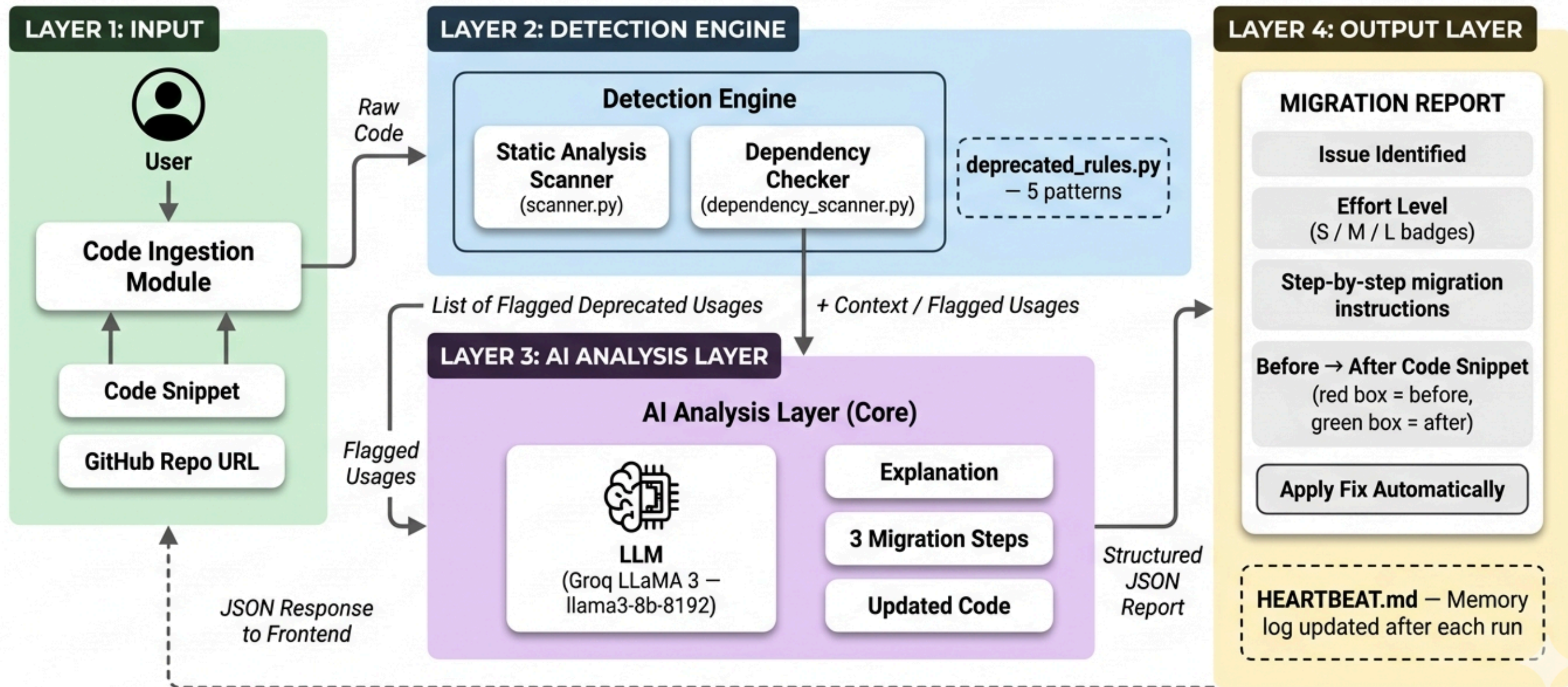
Key Features

- GitHub repository scanning and code ingestion
- Deprecated API and outdated dependency detection
- AI-generated migration explanations and upgrade steps
- Before → After code transformation view
- Structured JSON output for frontend rendering
- OpenClaw-style architecture using SOUL.md, SKILL.md, and HEARTBEAT.md

Demo/Product Walkthrough



DEBTMAP ARCHITECTURE (AI-Powered Migration Advisor)



Idea

USE CASES

STARTUP MOVING FAST

- Built AI chatbot in 2022 using `openai.Completion.create()` — deprecated in Jan 2024
 - App throws `AttributeError` in production
- DebtMap detects it, generates exact replacement code, fix applied in one click
- Time saved: 2–4 hours of reading OpenAI migration docs

ENTERPRISE LEGACY CODEBASE

- 5-year-old fintech backend using `os.system()` — flagged in security audit
 - Also uses `import cgi` — removed in Python 3.13
- DebtMap scans entire GitHub repo, flags 4 files, \$270 migration cost, effort: High
- Prevents security audit failure and Python upgrade breakage

OPEN SOURCE CONTRIBUTOR

- `package.json` depends on `request` — deprecated npm library (2020)
- DebtMap queries live npm registry, finds the deprecated flag, suggests `axios`
 - Developer fixes before their users start seeing warnings

SCALE POTENTIAL

- Today: 5 patterns, Python focused, single repo scan
- 3–6 months: 100+ rules, multi-language, CI/CD GitHub Action integration
- 6–18 months: VS Code extension, SaaS dashboard, auto-PR generation
- Long term: Enterprise tier, AI fine-tuned on real migration PRs, predictive deprecation alerts

Differentiation

AI Migration Intelligence

DebtMap detects deprecated APIs, understands semantic API changes, and generates migration steps automatically.

Tools like ESLint and Snyk mainly stop at static warnings and cannot explain migration paths.

One-Click Developer Workflow

DebtMap provides browser-based PR patching, GitHub integration, and one-click AI auto-fixes.

Most existing tools only identify problems and still require developers to manually search docs and rewrite code themselves.

Enterprise-Grade Insights

DebtMap estimates migration effort, tracks historical API evolution, and supports custom internal migration rules.

Traditional linters and dependency scanners do not provide migration intelligence or long-term technical debt visibility.

Demo Link

GitHub Repo : <https://github.com/Manaswini-Uppuluri/DebtMap>

Demo Link: <https://debtmap.onrender.com/>

Document submitted : AI Disclosure form, Demo Video, PPT